

Demonstration of 3D Printed Magnetophoretic Displays

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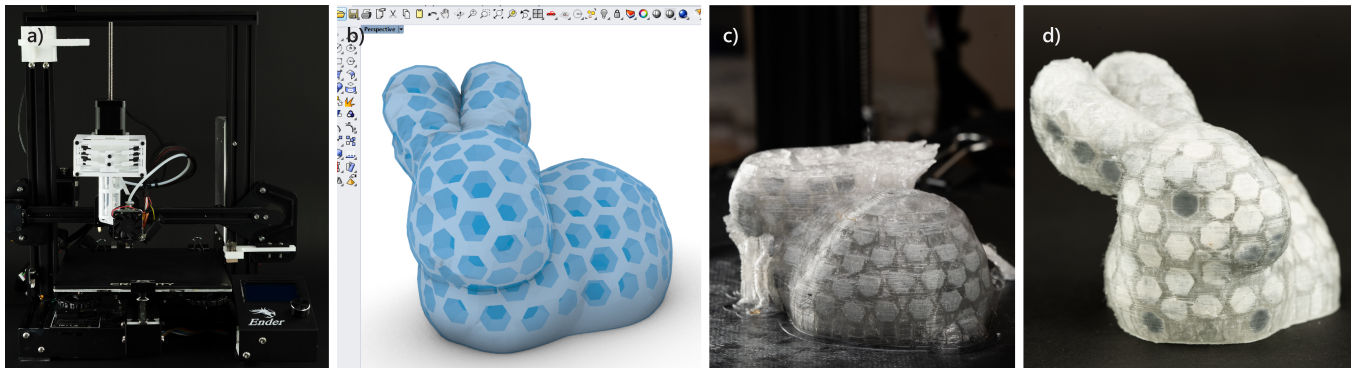


Figure 1: Printing pipeline overview. a) A modified FDM 3D printer with an additional syringe injector. b) A 3D editor that converts a model into a magnetophoretic display. c) Printing in-progress. d) The appearance of the printed model can be post edited.

ABSTRACT

In this demonstration, we showcase a suite of 3D printed magnetophoretic displays using the techniques proposed in [8]. These examples include a printed Stanford bunny with customizable appearances, a small espresso mug that can double as a post-it note surface, a board game figurine with a computationally updated display, and a collection of flexible wearable accessories with editable visuals. We will demonstrate the 3D printer design as well as the primary process of printing these magnetophoretic displays.

CCS CONCEPTS

• **Human-centered computing** → **Interaction devices; Systems and tools for interaction design.**

KEYWORDS

Magnetophoretic, 3D Printing Display, Low Power Display, Liquid Injection, 3D printing, 3D Printer Modification

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1 INTRODUCTION

Research advancements in 3D printing have enabled end users to design and fabricate a variety of dynamic 3D artifacts, ranging from deformable bespoke objects [3] to custom speakers [4]. However, printing 3D artifacts with appearances that are non-static or interactive remains challenging. Recent studies such as Printed Optics [7], PAPILLON [2], and computational light routing [5], suggest altering the appearance of a 3D printed artifact by embedding fiber optics or similar structures to change objects' appearances. These approaches require high printing resolution, which is only available from high-end inkjet or PolyJet printers.

This demonstration showcases an end-to-end pipeline that transforms arbitrary artifacts into self-contained, always-on, and interactive 3D surfaces using a modified consumer-grade fused deposition modeling (FDM) 3D printer. Our pipeline consists of a modified FDM 3D printer equipped with an additional syringe-type injector and a 3D design tool that converts a 3D model's surface into a matrix of magnetophoretic cells. We will present a set of application scenarios highlighting the customizability enabled by our pipeline.

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2 OVERVIEW OF THE PRINTING PRINCIPLE

A magnetophoretic display consists of a matrix of cells distributed across the display surface, with each cell containing a mixture of opaque liquid and magnetic powder; these two components typically exhibit contrasting colors.

Our pipeline focuses on facilitating the manufacturing of custom magnetophoretic displays. The automated manufacturing of the displays is made possible with a modified low-cost FDM desktop 3D printer (Creality Ender 3 Pro). We integrated a stepper-driven, syringe-based liquid injector next to the original FDM extruder to enable liquid injection. The liquid mixture comprises 1) iron powder, our choice of magnetic powder due to its high permeability, superior magnetic susceptibility [6] and wide accessibility [1], and 2) mineral oil, which serves as the liquid substrate to prevent the iron powder from rusting. The mixture is further thickened with talcum powder. Additional oil-based white dye is also added for better contrast.

The printing of magnetophoretic displays is similar to the conventional FDM process, where the printing materials are deposited layer by layer. When the print proceeds, certain cell(s) would have the majority of their volume printed, leaving an opening that fits the liquid injector nozzle.

We develop a 3D editor, a Grasshopper plugin for Rhino 3D programmed in C# with *Rhinocommon API*¹ and *Human UI*², enabling end-users to convert a 3D model into a magnetophoretic display. The plugin assists the user in generating distributed cells across the surface of a CAD model through a simple user interface. The user can modify cell parameters such as cell size and distribution gap, and the tool generates the updated model for G-code slicing. To ensure successful prints, any necessary manipulations to the G-code automatically proceed with a Python script when fed with the generated geometries from our tool. The custom G-code post-processor is developed with Trimesh³. With the modified G-Code, the entire printing process can be automated. All examples in the following section were printed using this pipeline.

3 EXAMPLES

In this section, we showcase examples to illustrate the potential and capabilities of our magnetophoretic display printing pipeline.

Stanford Bunny and Panton Chair. The main promise of our proposed pipeline is its ability to convert an arbitrary 3D artifact into a magnetophoretic display, the appearance of which can be post-edited. As an illustration, we showcase a Stanford bunny model printed using our pipeline. With our 3D editor, we converted the surface of the model into a pattern of hexagon voxels. Figure 2a presents a bunny model with post-decorated eyes, legs, and ears. In addition, we printed a scaled-down Panton chair to demonstrate a 3D display on a non-developable surface that can be viewed and decorated from both sides (Figure 2b).

Espresso Cup as Post-it. Post-it or sticky notes are commonly used to leave messages and reminders for ourselves and others. In this example, a round espresso cup has been transformed into a post-it cup. As shown in Figure 2c and d, the front of the cup

is converted and printed with a grid of 16 by 10 cells, effectively turning the curved surface into a drawing area.

Personalized Clothing. Clothing is a form of personal statement for many. Personalized designs have been used to make one-of-a-kind clothing items, such as T-shirts with custom slogans or logos. Here, we showcase two wearable accessories—a smartwatch strap and a handbag—printed using our pipeline. The watch strap is printed with elastic TPU so that it can be bent and fit the user’s wrist Figure 2e-f. The decorative pattern on the strap can be easily altered across the surface. Similarly, Figure 2g-h showcases a handbag assembled with hexagon tiles with surfaces being magnetophoretic displays. The handbag can be re-patterned with different slogans, enabling a fresh look to coordinate with various outfits.

Board Game Figurine. In this example, we demonstrate that a 3D-printed miniature can automatically update its appearance based on the progression of a board game by erasing existing patterns drawn on the figurine. In Figures 2j and k, certain patterns of a figurine are computationally removed, denoting various feature updates throughout a board game. The erasure of these patterns is achieved using a wireless motorized platform (Figure 2i). Its magnetically equipped end-effector can move along a cylindrical surface, erasing already-drawn patterns.

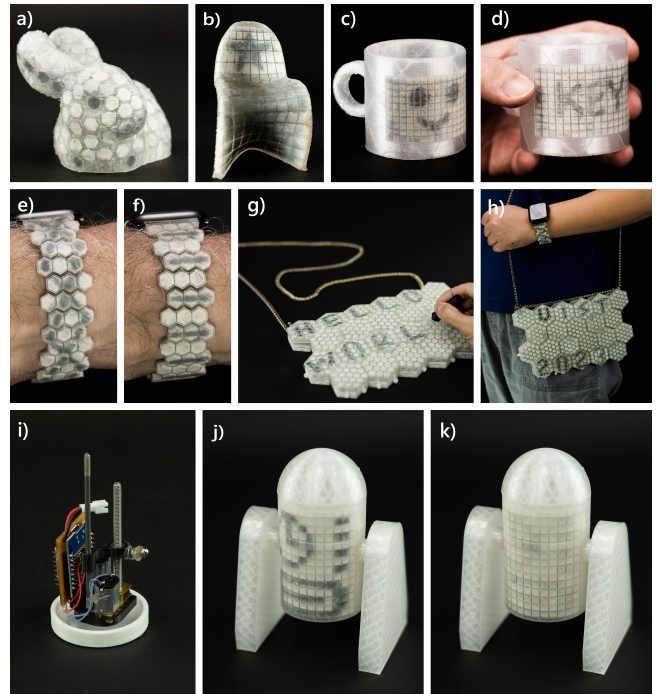


Figure 2: Examples.

4 CONCLUSION

Through a series of examples, this demonstration showcased the capability of a 3D printing pipeline for creating custom magnetophoretic 3D displays.

¹Rhinocommon API: <https://developer.rhino3d.com/>

²Human UI: <https://grasshopperdocs.com/addons/human-ui.html>

³Trimesh: <https://trimsh.org/index.html>

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